Use Cases

virtusa

Title	Description	Details
Smart Match	Precision Hiring with Skill Based Candidate Matching. To develop a User Interface (UI) that matches candidates with job descriptions (JDs) based on their skills, categorizes them, and generates reports.	> Automated Skill Extraction: Parses JDs and resumes to identify and weigh skills, ensuring accurate matches. > Candidate Categorization: Automatically classifies candidates as interns, freshers, or experienced based on their profiles > Match Scoring System: Compares candidate skills to JD requirements, assigning precise match scores. > Detailed Reporting: Generates insightful reports ranking candidates and JDs by skill weightage. > User Friendly Interface: Provides an intuitive web UI for easy management and decision making by HR staff.
Al Quiz	Intelligent MCQ Generation and Assessment Chatbot. To streamline and enhance the interview process by leveraging artificial intelligence to automatically generate and administer skill-based multiple-choice questions (MCQs) through a chatbot interface.	> Candidate Selection Interaction: Engages shortlisted candidates via an Al-powered chatbot. > Automated Profile Testing: Conducts skill assessments through interactive chatbot sessions. > Custom MCQ Generation: Creates tailored multiple-choice questions for relevant skills during the interview process. > Skill-Based Reporting: Produces detailed reports ranking candidates based on their skill performance. > Optional Face Capture: Ensures identity verification by capturing candidates' faces during mock tests.

AI Quiz: Intelligent MCQ Generation and Assessment Chatbot

Problem Statement

To streamline and enhance the interview process by leveraging AI to automatically generate and administer skill-based MCQs through a chatbot interface.

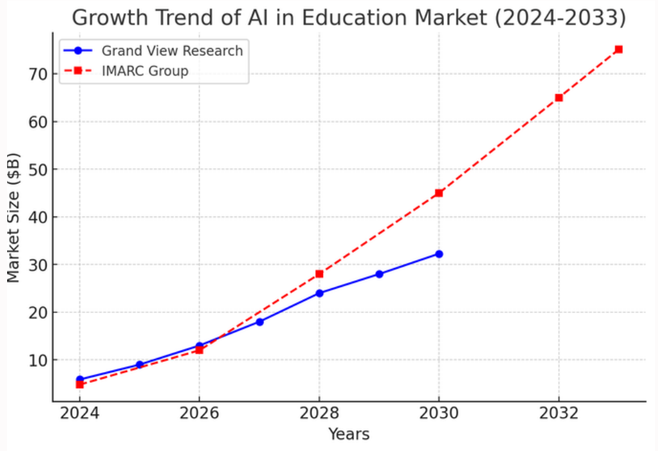
Business Need/Opportunity

- **Adaptive MCQs** via **RAG**, **SLM**, and **LLMs** for *real-time, context-aware question generation*
- **Multi-modal assessments** using **GNNs**, **Bayesian IRT**, and **CoT** *for deep skill analytics*
- **Blockchain verification** with **Ethereum/Hyperledger** smart contracts and **DIDs** for *tamper-proof USIDs*
- **Real-time proctoring** with *eye-tracking, facial recognition, and sentiment analysis*
- **Cloud-native, API-first microservices** for *scalable, secure integration*

Target Segment

- **Enterprises & Banking services** (HR & Recruitment)
- **Educational Institutions** (Universities, Online Learning Platforms)
- **Government & Military** (High-security & stress-sensitive exams)
- **Healthcare & Aviation** (Critical decision-making roles)

Market Size and Growth



31.2% CAGR (2025-2030) by Grand View and 34.03% CAGR (2025-2033) by IMARC

Future Readiness

1 — Research & Planning
Establish technical feasibility by conducting market research, competitor analysis, defining system requirements, finalizing core functionalities, and creating the technical architecture with UI/UX wireframing.

2 — MVP Development
Develop an MVP with a hyper-adaptive chatbot, AI-driven MCQ generation, a recruiter dashboard, skill-based ranking, AI cheating detection, and face recognition for identity verification.

3 — Beta Testing & Refinement
Beta test the MVP, gather feedback, refine AI for adaptive MCQs, optimize reports, enhance rankings, and finalize security and cheating detection.

4 — Full-Scale Development & Scalability
Enhance AI for stress detection in exams, expand IoT for physiological monitoring, refine real-time adaptive MCQ generation, upgrade the recruiter dashboard.

5 — Market Launch & Expansion
Launch the solution by deploying it on AWS or another cloud platform, executing an enterprise sales and marketing strategy for HR tech firms, conducting pilot programs with enterprises and universities, gathering post-launch feedback.

** RAG: Retrieval Augmented Generation; SLM: Skill-Level Modeling; LLM: Large Language Model; GNN: Graph Neural Network; IRT: Item Response Theory; CoT: Chain-of-Thought Prompting; DID: Decentralized Identifier; USID: Universal Skill ID; API: Application Programming Interface; CAGR: Compound Annual Growth Rate; MVP: Minimum Viable Product.*

Team Name:
LicenceToScript

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- **Harshita Talwar**
- **Ashish Bind**

Solution Details

- **Streamlined Interview Process** -AI based Skill mapping, Skill-Based Question Generation & Reporting
- **Automated Administration** - Hierarchical skill report before, during and after the interview
- **Scalable + Cost-Effective Solution:** Leveraging the power of Microservice arch. and Kubernetes

Unique Selling Proposition

01

Dynamic Skill - Gap Mapping:

- **GNN-** Map resume vs job description skill gap
- **Hierarchical Architecture** - of required job-skills(most imp to least imp) and of AI generated skill-match report

02

HyperAdaptive Chatbot Environment

- **SLM, PLM -** Real time difficulty level personalisation(based on skill-level and personalized learning) via Multi-Model arch.
- **RAG-** Contextually Aware Questions

- *Automated Profile Testing*
- *Custom MCQ Generation*

03

Key - Features:

- **Multi-Modal Model-** Handle video+textual data, process the data on PLM, SLM, LLM+RAG
- **Emotional AI-** Cheating Detection for high stake exams
- **Blockchain Based Reporting-** Trustworthiness of Reports

04

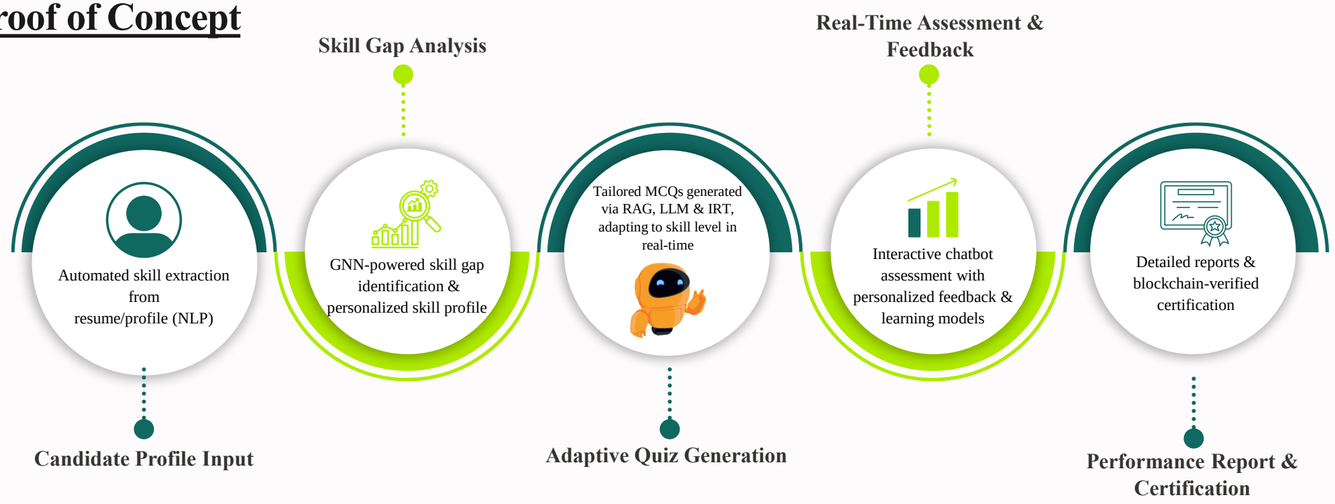
Benefits:

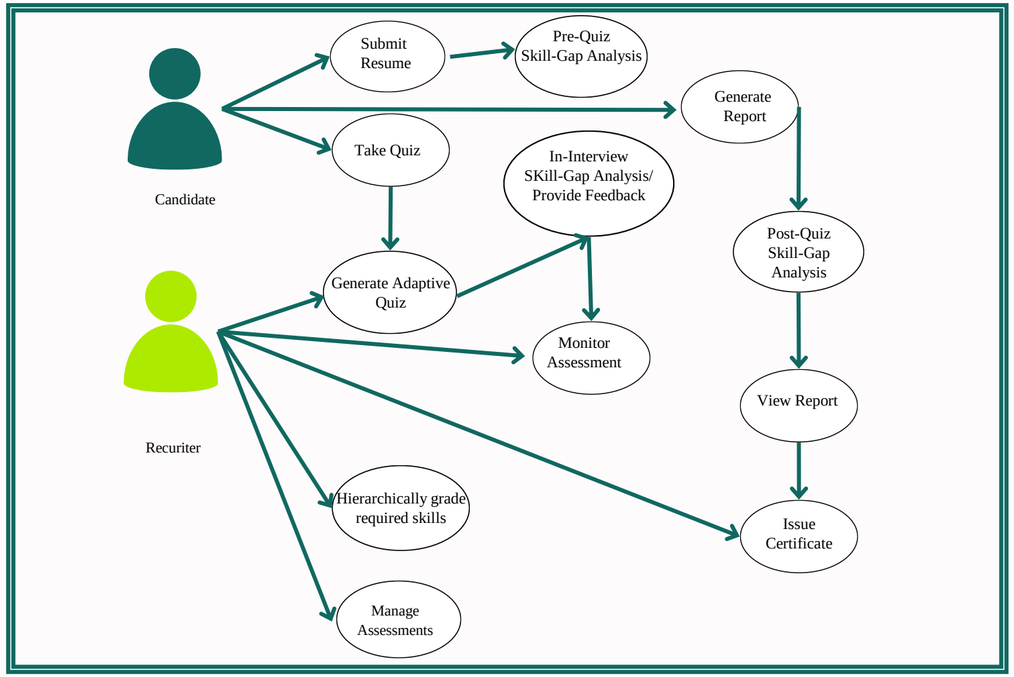
- **Cost-Effective, Scalable, Faster Market Time-** Microservice Architecture along with Kubernetes.
- **Enhanced UX-** Gamified Environment, Intuitive Chatbot

- *Key features and unique benefits*

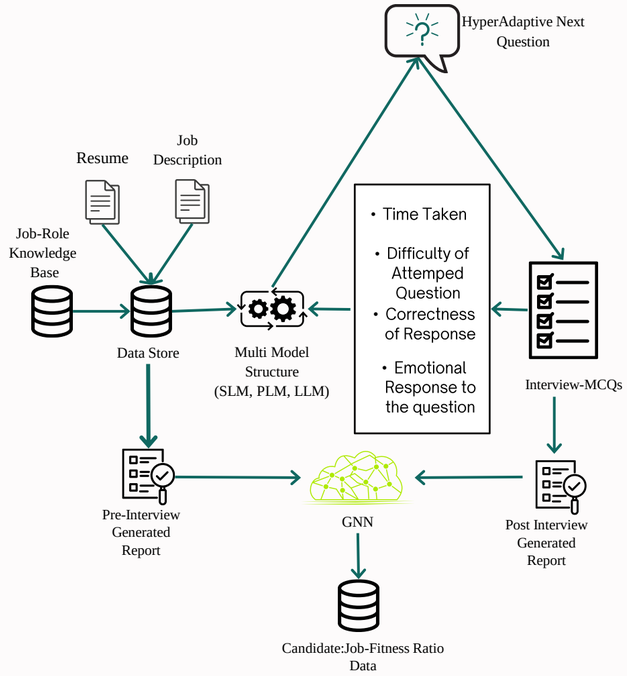
**PLM: Personalized Learning Model*

Proof of Concept





- ***Use Case Diagram***



- **Working**